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3 Mathematical Introduction to Dynamical Systems

No roote, no fruite

Jeremiah Dyke, A worthy communicant,

(1640)

In this chapter, we present in the simplest possible manner a survey of some of the fundamental mathematical concepts and tools which are required for the qualitative analysis of the long-term behaviour of dynamical systems. A knowledge of the theory of linear differential equations is a pre-requisite for the comprehension of non-linear dynamics. The reader can find more detailed discussions in Chapters 5 and 6 of this book.

3.1 Linear Autonomous Systems

As we already mentioned in Chapter 2, we consider dynamical processes of a finite dimension n which are differentiable and deterministic.

In the simplest case, the system of differential equations (2.3.3) is linear with constant coefficients

$$\dot{\mathbf{x}} = \mathbf{L}\mathbf{x}$$

where $\mathbf{L}$ denotes the constant, non-singular $(n \times n)$ -matrix of the coefficients. Using this example, we now demonstrate which solutions can occur and which patterns the relevant trajectories form in the corresponding phase space.

We introduce the ansatz

$$\mathbf{x} = e^{\lambda t}\mathbf{y}, \quad \mathbf{y} = \{y_1, y_2, \dots, y_n\}$$

into eq. (3.1.1), thus obtaining a homogeneous linear system of equations

$$[\mathbf{L} - \lambda\mathbf{I}]\mathbf{y} = 0$$

which only possesses non-trivial solutions if the system determinant vanishes

$$P(\lambda) = \det[\mathbf{L} - \lambda\mathbf{I}] = 0$$

or

$$P(\lambda) = A_n\lambda^n + A_{n-1}\lambda^{n-1} + \dots + a_0 = 0$$

$P(\lambda)$ is a polynomial of the n th degree in λ and is called characteristic or secular equation. The zeroes of $P(\lambda)$ are the eigenvalues of $\mathbf{L}$. A non-vanishing vector $\mathbf{y}$

which satisfies eq. (3.1.3) is called the eigenvector of $\mathbf{L}$ appertaining to the eigenvalue λ . If λ and $\mathbf{y}$ satisfy eq. (3.1.3), eq. (3.1.2) is a solution of eq. (3.1.1). For each pair λ_i, y_i , we obtain in accordance with eq. (3.1.2) a solution of the form

$$x_i = e^{\lambda_i t} y_i$$

If all n eigenvalues λ_i are different, n linearly independent eigenvectors y_i exist and the general solution of eq. (3.1.1) can be expressed by the linear combination

$$x(t) = \sum_{i=1}^n C_i e^{\lambda_i t} y_i$$

with n integration constants C_i which are determined by the initial conditions $x(t_0)$ (Arnold, 1980; Braun, 1979).

From a physical point of view, finding all the states of equilibrium x_s is of interest. These are stationary states of the system in which the dynamical process does not undergo any change; they are thus characterised by $\dot{x}_s = 0$. If we observe the corresponding point in the phase space, the vector $F(x_s)$ which defines the change of

the trajectory vanishes here. For this reason, x_s is thus also denoted singular point. The decisive factor in characterising the stationary state is the behaviour of the trajectories in the neighbourhood of the singularity. We speak of stable, unstable or neutral equilibrium. If all the trajectories within certain neighbourhoods of x_s are captured, the singular point is asymptotically stable (sink); if, on the other hand, all the trajectories that come

close enough to $\mathbf{x}_s$ are repelled, $\mathbf{x}_s$ is asymptotically unstable (source). On the basis of a system of two differential equations, we now demonstrate the classification of singularities

$$\dot{\mathbf{x}} = \mathbf{L}\mathbf{x}, \quad \mathbf{L} = \begin{bmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{bmatrix}, \quad \mathbf{x} \in \{x_1, x_2\}$$

with the origin $\mathbf{x}_s = 0$ as the singular point. The eigenvalues of the corresponding eq. (3.1.3) serve as a basis to make the following six distinctions, illustrated in the diagram fig. 3.1.1; here, we ignore the degenerate case of a zero eigenvalue.

$$(A) \quad \lambda_1 \neq \lambda_2$$

Initially, we assume that $\mathbf{L}$ possesses different eigenvalues $\lambda_1 \neq \lambda_2$ and denote the corresponding eigenvectors as $\mathbf{y}_1, \mathbf{y}_2$ which we can presume to be normalised without loss of generality.

We now introduce a new system of coordinates

$$\bar{\mathbf{x}} = \{\bar{x}_1, \bar{x}_2\}$$

which is defined by the transformation

$$\mathbf{x} = \mathbf{T}\bar{\mathbf{x}} \text{ with } \mathbf{T} = [\mathbf{y}_1 \mathbf{y}_2]$$

Thus, the eigenvectors $\mathbf{y}_1, \mathbf{y}_2$ are the basis vectors of the new coordinate system. Substitution of the transformation eq.(3.1.10) into the given system eq.(3.1.8) yields

$$\dot{\bar{\mathbf{x}}} = \mathbf{D}\bar{\mathbf{x}} \text{ with } \mathbf{D} = \mathbf{T}^{-1}\mathbf{L}\mathbf{T}$$

Due to

$$\lambda_i \mathbf{y}_i = \mathbf{L}\mathbf{y}_i$$

a

$\mathbf{D}$ can be simplified as follows

$$\begin{aligned} D &= [\mathbf{y}_1 \ \mathbf{y}_2]^{-1} \mathbf{L} [\mathbf{y}_1 \ \mathbf{y}_2] = [\mathbf{y}_1 \ \mathbf{y}_2]^{-1} [\mathbf{y}_1 \ \mathbf{y}_2] [\lambda_1 \ \lambda_2] \\ &= [\lambda_1 \ \lambda_2] \end{aligned}$$

and is thus reduced to diagonal form. The transformed system equivalent to eq. (3.1.8) now becomes

$$\begin{aligned}\dot{\bar{x}}_1 &= \lambda_1 \bar{x}_1 \\ \dot{\bar{x}}_2 &= \lambda_2 \bar{x}_2\end{aligned}$$

Here, we can distinguish for $\lambda_1 \neq \lambda_2$ a total of four possibilities (see fig. 3.1.1). If the eigenvalues are real, the solution of eq. (3.1.13) according to eq. (3.1.7) is

$$\bar{x}_1 = C_1 e^{\lambda_1 t}, \quad \bar{x}_2 = C_2 e^{\lambda_2 t}$$

or, after eliminating the time parameter t ,

$$\bar{x}_2 = C \bar{x}_1^{\lambda_2/\lambda_1}$$

We next differentiate between two cases having different signs (see fig. 3.1.1);

$$(A1) \quad \text{sign} \lambda_1 = \text{sign} \lambda_2$$

If the eigenvalues have the same sign, eq. (3.1.15) describes a family of parabolae of the order λ_2/λ_1 which have a common tangent at their origin (see fig. 3.1.2). We then speak of a node. If the eigenvalues are negative, the node is stable, as can be seen in the parameter representation of eq. (3.1.14) since all trajectories tend towards the singular point for $t \rightarrow \infty$.

$$(A1) \quad \text{sign} \lambda_1 = \text{sign} \lambda_2$$

If λ_1 and λ_2 possess different signs, the solution is

$$\bar{x}_2 \bar{x}_1^{-|\lambda_2/\lambda_1|} = C$$

i.e. the trajectories are hyperbolae. The corresponding singular point is then called a saddle point or hyperbolic point (see fig. 3.1.3).

If λ_1, λ_2 are complex conjugate, $\mathbf{L}$ can always be reduced by applying a (real) linear coordinate transformation $\hat{T}$ to the simple form

$$\dot{\hat{\mathbf{x}}} = \hat{L} \hat{x} \quad \text{with} \quad \hat{L} = \begin{bmatrix} \alpha & -w \\ w & \alpha \end{bmatrix}$$

The eigenvectors $\mathbf{y}_1$ and $\mathbf{y}_2$ appertaining to the eigenvalues $\lambda_{1,2} = \alpha \pm iw$ are also complex conjugate

$$\mathbf{y}_1 = \{1 - i\} \text{ and } \mathbf{y}_2 = \{1i\} = \mathbf{y}_1^*$$

and reduce in accordance with eq. (3.1.10) the transformation of $\hat{L}$ to a diagonal form; in accordance with standard usage, $\mathbf{y}_1^*$ denotes the complex conjugate vector of $\mathbf{y}_1$. Thus, points $\hat{x}$ of the real phase space are transformed into points $\bar{x}$ of a two-dimensional complex phase space $\mathbf{C}^2$. The solution of eq. (3.1.13) can be expressed immediately as

$$\bar{x}_{1,2} = C_{1,2}e^{(\alpha \pm iw)t} = C_{1,2}e^{\alpha t}(\cos wt \pm i \sin wt)$$

where the complex constants C_1 and C_2 coincide with the initial values $\bar{x}_1(0)$ and $\bar{x}_2$. It is now easy to see that C_1 and C_2 are complex conjugate. From the inverse transformation of eq. (3.1.10) for an initial vector $\hat{\mathbf{s}}(0)$

$$\bar{\mathbf{x}}(0) = \mathbf{T}^{-1}\hat{\mathbf{x}}(0) = [\mathbf{y}_1 \quad \mathbf{y}_1^*]^{-1}\hat{\mathbf{x}}(0) = \frac{1}{2} \begin{bmatrix} \mathbf{y}_1^{*t} \\ \mathbf{y}_1^t \end{bmatrix} \hat{\mathbf{x}}(0)$$

we obtain the relation $\bar{x}_2(0) = \bar{x}_1^*(0)$, i.e. $C_2 = C_1^*$. Thus, the solutions according to eq. (3.1.17) are also complex conjugate, i.e. $\bar{x}_2(t) = \bar{x}_1^*(t)$ applies. The inverse transformation of the initial system hence yields a real solution

$$\hat{x}(t) = e^{\alpha t} \begin{bmatrix} \cos wt & -\sin wt \\ \sin wt & \cos wt \end{bmatrix} \hat{x}(0)$$

We observe that the initial vector $\hat{x}(0)$ is rotated about an angle wt and stretched with the factor $e^{\alpha t}$.

We once again distinguish between two cases (see fig. 3.1.1):

$$(\mathbf{A3}) \alpha = \text{Re}(\alpha_i) \neq 0$$

Equation (3.1.18) suggests a representation in polar coordinates for which we set

$$r = r_0 e^{\alpha t}, \varphi = wt + \varphi_0$$

with $r_0 = |\hat{x}(0)|$ and $\varphi_0 = \tan^{-1}(\hat{x}_2(0)/\hat{x}_1(0))$. This formulation describes a family of logarithmic spirals. In accordance with an Anglo-Saxon convention, we use the inverse function sign for arcus. If the real part of λ_i is negative, i.e. $Re(\lambda_i) = \alpha < 0$, we have a stable focus or spiral point; on the other hand, for $\alpha > 0$, the singular point is unstable. Figure 3.1.4 shows the spirals in the original x_1, x_2 coordinate system. They are affinely distorted by the inverse transformation.

$$(\mathbf{A4}) \alpha = Re(\lambda_i) = 0$$

In the special case of $\alpha = 0$, the phase curves in the $\hat{x}_1, \hat{x}_2$ coordinate system become concentric circles in accordance with eq. (3.1.18), the centre of which is the singular point. We then speak of a centre. Figure 3.1.5 shows the trajectories in the original x_1, x_2 coordinate system in which $\mathbf{L}$ possesses a general form. Here, we observe concentric ellipses.

$$(\mathbf{B}) \lambda_1 = \lambda_2 = \lambda$$

We now move on to a discussion of the case of identical eigenvalues, $\lambda_1 = \lambda_2 = \lambda$. Here, we distinguish between two cases (fig. 3.1.1):

(B1) There are two linearly independent eigenvectors $\mathbf{y}_1, \mathbf{y}_2$ for λ .

In analogy to eqs. (3.1.11), (3.1.12), the system $\dot{\mathbf{x}} = \mathbf{L}\mathbf{x}$ can be transformed into a diagonal form, $\dot{\bar{\mathbf{x}}} = \lambda \bar{\mathbf{x}}$ ($i = 1, 2$), with the solutions

$$\bar{x}_i = C_i e^{\lambda t} (i = 1, 2)$$

or

$$\bar{x}_2 = C \bar{x}_1$$

We thus obtain a bundle of straight lines and speak of a dicritical node (see fig. 3.1.6).

In this case, a sufficient number of eigenvectors does not exist to span the whole phase space; thus, $\mathbf{L}$ can no longer be transformed to diagonal form. However, in matrix theory, it is shown that any non-diagonalisable ($n \times n$) matrix can always be transformed uniquely to the so-called Jordan normal form where each Jordan block is associated with an eigenvector (Zurmühl, 1964).

For $n = 2$, there is thus a specific $\bar{x}_1, \bar{x}_2$ coordinate system in which the system of differential equations (3.1.8) can be transformed to the Jordan normal form

$$\dot{\bar{\mathbf{x}}} = \bar{\mathbf{L}}\bar{\mathbf{x}} \text{ with } \bar{\mathbf{L}} = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$$

The corresponding coordinate transformation is

(B2) There is only one linearly independent eigenvector $\mathbf{y}$ for λ .

$\mathbf{x} = \mathbf{T}^{-1}\bar{\mathbf{x}}$ with $\mathbf{T} = [\mathbf{y} \ \mathbf{z}]$ (3.1.23) where a second vector $\mathbf{z}$ linearly independent of $\mathbf{y}$ must be established. This vector $\mathbf{z}$ is denoted generalised eigenvector. Pre-multiplying $-\mathbf{L} \ \mathbf{L} = \mathbf{T}^{-1}\mathbf{L}\mathbf{T} \ \mathbf{L}\mathbf{T}$ with $\mathbf{T}$ yields $\mathbf{T}^{-1}\mathbf{L}\mathbf{L} = \mathbf{L}\mathbf{T}$. Evaluating the matrix products, we find $\mathbf{T}^{-1}\mathbf{L}\mathbf{L} = [\mathbf{y} \ \mathbf{z}] \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = [\mathbf{y} \ (\mathbf{y} + \mathbf{z})]$ and $\mathbf{L}\mathbf{T} = \mathbf{L}[\mathbf{y} \ \mathbf{z}] = [\mathbf{y} \ \mathbf{L}\mathbf{z}]$. Comparing the second columns and resolving for $\mathbf{y}$ yields $[\mathbf{L} \ \mathbf{I}]\mathbf{z} = \mathbf{y}$.